

Chansoo Son

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Education & Military Service

Gachon University Department of AI&Software (2024.3~2026.8)

- Major GPA: 3.91 / 4.5
- Overall GPA: 3.75 / 4.5

Gachon University, Department of AI Software, Artificial Intelligence Major, RA (2024.8 ~ 2026.2)

- Intelligent Robotics and Autonomous System Control Lab (iRASC)

Honorable Discharge from the Army as a Sergeant (2022.1 ~ 2023.7)

- Specialized Skilled Soldier – Drone Operation and Maintenance

Research Interests

- Multi-Agent Reinforcement Learning (MARL): Decision Making & Control, Cooperative/Competitive Scenarios, Multi-Drone Systems
- Problem Formulation & Evaluation: MDP Design (State/Action/Transition/Observation), Reward Engineering, Robust Evaluation Protocols
- MARL Stability & Generalization: Non-stationarity, Self-play Mode Collapse/Cycling, POMDPs
- Applications: Manned-Unmanned Teaming (MUM-T), Space Robotics Control, Autonomous Navigation

Research Experience

- Methodology Research: Stability & Generalization in MARL
 - Formulated robust Markov Decision Processes (MDP) and learnable reward/evaluation protocols for cooperative and competitive multi-drone environments.
 - Addressed critical MARL challenges, including non-stationarity from shifting opponent policies and mode collapse/cycling in self-play.
 - Developed reliable policy learning strategies for environments with limited sensors and partial observability (POMDP).
 - Established standardized learning procedures and evaluation protocols to ensure high reproducibility and comparability across MARL experiments.
- LLM-based Manned-Unmanned Teaming (MUM-T) System
 - Led the UAV decision-making component within an LLM-driven MUM-T framework.
 - Integrated natural language mission instructions with autonomous UAV behaviors, bridging the gap through automated mission decomposition, action selection, and real-time situational awareness.
- Space Robotic Arm Docking Control via PPO
 - Engineered a Proximal Policy Optimization (PPO)-based control policy for robotic arm docking in a simulated space environment.
 - Modeled complex control objectives into RL frameworks to achieve stable approach, precise alignment, and contact maintenance under highly constrained sensor and contact conditions.

Publications(Domestic)

- Taewan Cho, **Chansoo Son**, Mingyu Jung, Hyunwoo Yoo, Changeon Park, Jaeyong Choi, "Development Trends in Flight Safety Verification Methods for AI-based Military Aircraft Embedded Software," The Korean Society for Aeronautical and Space Sciences (KSAS), Jun 2025.

Conferences(International)

- **Chansoo Son** and Andrew Jaeyong Choi, "Beyond the Ring: A Multi-Agent Reinforcement Learning Framework for Attack and Defense Intelligence in Drone Soccer" 2nd International Conference on Drones and Unmanned Systems (DAUS' 2026), (2026. 2. 25~27)

Conferences(Domestic)

- **Chansoo Son**, Jaeyong Choi, "ADS-RL: Autonomous Drone Soccer using Deep Reinforcement Learning," KSAS Fall Conference, Delpino Resort, Goseong, Gangwon-do, Nov 11-24, 2025.
- Suhyun Lim, Yubin Shin, **Chansoo Son**, Jaeyong Choi, "6-DOF Drone Control System using Graph-based Single-hand Gesture Recognition," KSAS Fall Conference, Delpino Resort, Goseong, Gangwon-do, Nov 11-24, 2025.
- **Chansoo Son**, Jaeyong Choi, "Autonomous Drone Soccer using Deep Reinforcement Learning," IEIE Summer Conference, Lotte Hotel Jeju, Jeju, Jun 24-29, 2025.
- Taewan Cho, **Chansoo Son**, Hyunwoo Yoo, Rina Kim, Jaeyong Choi, "A Study on Multimodal LLMs for Intelligent Multi-Robot Systems," IEIE Fall Conference, High1 Resort Convention Tower, Jeongseon, Gangwon-do, Nov 22-23, 2024.

Projects

- A Study on Flight Safety Verification Methods for AI-based Military Aircraft Embedded Software
 - Period: Jun 2024 - Nov 2024
 - Client: Defense Agency for Technology and Quality (DTaQ)
 - Main Objective: To derive airworthiness certification methods for AI (Software).
 - Key Contributions:
 - Investigated and compared domestic and international standards/guidelines regarding airworthiness and safety certification for AI and embedded software in military aircraft, focusing on scope, required evidence, and evaluation procedures.
 - Analyzed the differences between traditional SW certification systems and AI-embedded SW (e.g., data dependency, modifiability of learning-based models, operational environment variances) and identified core risk factors from a certification perspective.
 - Structured a functional decomposition (functions/requirements/operational scenarios/failure modes) to integrate embedded AI functions into the system-level safety analysis framework and established risk-based verification points.
- A Study on the Current Status and Fostering Plan for the Drone Leisure Industry
 - Period: Jun 2024 – Dec 2024
 - Client: Ministry of Land, Infrastructure and Transport (MOLIT) - Advanced Aviation Division
 - Main Objective: Expanding drone parks for hobby/leisure purposes, internationalizing drone sports (soccer, racing, etc.), and developing diverse drone e-sports.
 - Key Contributions:

- Surveyed domestic and international trends in drone leisure and sports. Categorized entry barriers (safety, cost, space, regulations) and expansion prerequisites (infrastructure, league/tournament structures, rules, education systems) by event type.
- Derived policy ideas for expanding drone parks and dedicated spaces, focusing on safety management (operational regulations, flight zone setups, spectator routes), user experience (beginner to advanced course designs), and regional connection programs.
- Analyzed rule design elements (match duration, scoring systems, penalties, collision/contact rules, equipment constraints) to enhance fairness and spectator engagement, proposing improvements for the popularization and internationalization of drone sports.

Awards

- Best Paper Presentation Award, 2025 KSAS Fall Conference
 - Designed a single-hand gesture control system for 6-DOF drone maneuvering and built an embedded control system using Raspberry Pi.
- LIG Nex1 Capstone Design Award, 2024 6th National Undergraduate Capstone Design Contest
 - Developed a UAV control algorithm and devised a UGV path planning system using Path Planning algorithms within a MUM-T (Manned-Unmanned Teaming) environment.
- Grand Prize (Team 7), 2024 AI Software Department P-Practical Project
 - Developed a real-time object detection model utilizing YOLO v11, achieving over 95% accuracy in detecting 13 types of vegetables.
 - Demonstrated proficiency in mobile AI vision services by quantizing and optimizing the model for smartphone camera environments using TensorFlow Lite.
- 1st Place (Commander of Ground Operations Command Award), 2023 Yangju Dronebot Festival - Drone Battle Category
- 2nd Place (Minister of National Defense Award), 2022 Minister of National Defense Dronebot Contest - Drone Soccer Category
- 2nd Place (Chief of Staff of the Air Force Award), 2022 6th Chief of Staff of the Air Force Drone Contest - National Drone Soccer Competition
- 2nd Place (Commander of Ground Operations Command Award), 2022 Yangju Dronebot Festival - Drone Battle Category
- 2nd Place, 2022 5th Segye Ilbo National Drone Soccer Competition
- 2nd Place, 2021 Ulsan Drone Mission Contest - National Drone Soccer Competition
- 2nd Place, 2021 2nd Korea University Drone Soccer Competition
- 3rd Place, 2021 WCRC 2nd Finals - Logistics Robot 2 (Sorting Category)

Skills and Techniques

- Programming Languages
 - Python, C
- Frameworks & Libraries
 - ROS1, ROS2, PyTorch, Skrl, Gymnasium, Stable baseline3, Rsl_rl
- Simulators & Development Tools
 - Isaac Sim & Lab, Gazebo, Docker

- Flight Control Stack
 - PX4, MAVLink, QGC

Teaching Experience

- Teaching Assistant, Drones and Robotics | Dept. of Software, Gachon University
 - Fall 2025 (Prof. Jaeyong Choi)
 - Created guidelines for establishing a ROS 2 environment using Docker and provided TurtleBot examples.
 - Conducted 3 separate TA session lectures.
 - Spring 2025 (Prof. Jaeyong Choi)
 - Covered drone components, 6-DOF flight dynamics, and simulation utilization.
 - Conducted 1 TA session lecture.

Certifications

- Ultralight Flying Vehicle Pilot Certificate (Unmanned Multicopter - Type 1) | Korea Transportation Safety Authority (TS)